Reading

PREPARING TO READ

A **BUILD VOCABULARY** The words in blue are used in the reading passage. Read the text below. Then write the correct form of each word next to its definition.

Science has shown us that what we eat affects our **life spans**. But when it comes to food, most of us fixate on quantity. People often diet unsustainably, skipping too many meals or eating much less than they should. Unfortunately, such harsh dietary **restrictions** often end up having an adverse effect on our weight and, by extension, our health and **longevity**.

The human body is equipped with biological **mechanisms** that send signals to the brain whenever it needs food. "Overhunger," however, tells our brains that there is a food shortage and that we need to eat as much as possible. **Ultimately**, we end up eating excessively—much more than we would have over the next few meals.

For most of us, it is better to simply listen to our bodies and eat when we're hungry. However, the trick is knowing what to eat and when to stop. Generally, we want to minimize our consumption of fatty foods and simple carbohydrates, like those found in white bread. Instead, we should eat more proteins, fruit, and vegetables. It is worth noting, though, that there is no perfect **ratio** of food groups: what's healthy depends largely on lifestyle and **genetic** makeup. So try documenting what you eat and learning what works for you.

Finally, don't just listen to your body when you are hungry. Recognize when you don't actually need to eat, and stop once you've had enough.

1. ________________ (n) something that limits what we can do
2. ________________ (adj) biologically determined through genes or heredity
3. ________________ (n) the amount of time that something is alive for
4. ________________ (n) the ability to live for a long time
5. ________________ (adv) in the end
6. ________________ (n) a set of numbers that describes how things are related proportionally
7. ________________ (n) a system, or a well-established process

B **BUILD VOCABULARY** Complete the sentences with the correct form of the words in blue. Use a dictionary to help you.

component (n)	**contradictory** (adj)	**dismiss** (v)	**historically** (adv)
implication (n)	**outcome** (n)	**outnumber** (v)	**reconstruct** (v)

1. He's not clear about his own position. He makes ________________ statements throughout his research paper.
2. We don't have all the information, but we've tried our best to ________________ events using what we know.

3. They're probably going to lose because the other team ________ them three to one.

4. Their job performance has been unsatisfactory, so the company has little choice but to ________ them.

5. That may be the case today, but ________, that hasn't always been true.

6. The discovery may not seem like a big deal to most, but its ________ are profound.

7. The experiment led to ________ that were highly unexpected.

8. Diet is just one of many ________ of a healthy lifestyle.

C **USE VOCABULARY** Discuss these questions with a partner.

1. Have you ever come across health tips that are **contradictory**? Give examples.
2. What are some traits that are **genetic**?
3. What are some **implications** of longer **life spans**?

D **PREDICT** Note answers to the questions below. Then discuss with a partner.

Critical Thinking

1. What are some factors that can affect a person's life expectancy?

____

2. What do you think are the biggest factors that affect life expectancy?

____

E **SKIM** Skim the reading passage. Are any of your ideas in Exercise D mentioned? What other factors are mentioned that can affect a person's life expectancy?

A farmers' market in Riga, Latvia

DEVELOPING READING SKILLS

READING SKILL Asking Questions as You Read

As you read, note questions you have, and try to predict their answers. This can help you stay engaged and give you a deeper understanding of the text. For example, after reading paragraph A of the reading passage, you might ask, "Why are Passarino and Berardelli going to Molochio?" You might then predict that they want to talk to a centenarian. This is later confirmed in paragraph B.

Most importantly, asking questions can help you identify gaps or assumptions in the passage. For example, in paragraph C, Salvatore Caruso attributes his longevity to eating mostly figs and beans while growing up, and hardly ever any red meat. You might ask yourself upon reading this, "Is there any scientific basis for this?"

After you read, it's possible for some of your questions to remain unanswered. This could be because the writer has made assumptions or left out some information. It could also be that your question is not very relevant. Reassess your questions, and decide if they are questions that really need to be answered.

A **ASK QUESTIONS AS YOU READ** Read paragraphs A–H of the reading passage. As you read, note down questions you have in the chart. As you continue reading, note down if your questions are answered later in the passage, and if so, how.

Paragraph	Questions raised	Were they answered? If so, how?

B **ANALYZE** Work with a partner. Look at your questions in Exercise A.

1. Which of your unanswered questions were you able to infer answers for?
2. Did any of your questions reveal gaps or assumptions in the passage?
3. Were any of your unanswered questions not very relevant or important?

C **ASK QUESTIONS AS YOU READ** As you read the rest of the reading passage, note down questions you have. Check if they are answered in the paragraphs that follow.

Beyond 100

by Stephen S. Hall

A man from the province of Ogliastra in Italy—an area known for the longevity of its people—prepares food for his friends.

Our genes harbor many secrets to a long and healthy life. And now scientists are beginning to uncover them.

A On a crisp January morning, with snow topping the distant Aspromonte mountains and oranges ripening on the nearby trees, Giuseppe Passarino guided his silver minivan up a curving mountain road into the hinterlands of Calabria, mainland Italy's southernmost region. As the road climbed through fruit and olive groves, Passarino—a geneticist at the University of Calabria—chatted with his colleague Maurizio Berardelli, a geriatrician.[1] They were headed for the small village of Molochio, which had the distinction of numbering four centenarians[2]— and four 99-year-olds—among its 2,000 inhabitants.

B Soon after, they found Salvatore Caruso warming his 106-year-old bones in front of a roaring fire in his home on the outskirts of the town. Known in local dialect as "U' Raggiuneri, the Accountant," Caruso was calmly reading an article about the end of the world in an Italian version of a supermarket tabloid.[3] A framed copy of his birth record, dated November 2, 1905, stood on the fireplace mantle.

C Caruso told the researchers he was in good health, and his memory seemed prodigiously intact. He recalled the death of his father in 1913, when Salvatore was a schoolboy; how his mother and brother had nearly died during the great influenza pandemic of 1918–19; and how he'd been **dismissed** from his army unit in 1925 after accidentally falling and breaking his leg in two places. When Berardelli leaned forward and asked Caruso how he had achieved his remarkable **longevity**, the centenarian said with an impish smile, "No bacco, no tabacco, no venere—No drinking, no smoking, no women." He added that he'd eaten mostly figs and beans while growing up and hardly ever any red meat.

D Passarino and Berardelli heard much the same story from 103-year-old Domenico Romeo—who described his diet as "poco, ma tutto—a little bit, but of everything"—and 104-year-old Maria Rosa Caruso, who, despite failing health, regaled[4] her visitors with a lively version of a song about the local patron saint.[5]

E On the ride back to the laboratory in Cosenza, Berardelli remarked, "They often say they prefer to eat only fruit and vegetables."

F "They preferred fruit and vegetables," Passarino said drily, "because that's all they had."

G Although eating sparingly may have been less a choice than an involuntary circumstance of poverty in places like early 20th-century Calabria, decades of research have suggested that a severely restricted diet is connected to a long **life span**. Lately, however, this theory has fallen on hard scientific times. Several recent studies have undermined the link between longevity and caloric **restriction**.

H In any case, Passarino was more interested in the centenarians themselves than in what they had eaten during their lifetimes. In a field **historically** marred[6]

[1] A **geriatrician** is a doctor who specializes in old age.
[2] A **centenarian** is someone aged 100 or older.
[3] A **tabloid** is a newspaper that that is considered less serious than regular newspapers.
[4] If you **regale** people with songs or stories, you entertain them.
[5] A **patron saint** is a person from the past who is regarded as holy, and who is considered a protector or helper of a place or group of people.
[6] If something is **marred**, it is spoiled or damaged.

by exaggerated claims, scientists studying longevity have begun using powerful genomic technologies, basic molecular research, and—most importantly—data on small, genetically isolated communities of people to gain increased insight into the maladies[7] of old age. In regions around the world, studies are turning up molecules and chemical pathways that may **ultimately** help everyone reach an advanced age in good, or even vibrant, health.

I In Calabria, the hunt for hidden molecules and **mechanisms** that confer longevity on people like Salvatore Caruso begins in places like the *Ufficio Anagrafe Stato Civile* (Civil Registry Office) in the medieval village of Luzzi. The office windows here offer stunning views of snow-covered mountains to the north, but to a population geneticist the truly breathtaking sights are hidden inside the tall file cabinets ringing the room, and on shelf after shelf of precious ledgers numbered by year—starting in 1866. Despite its well-earned reputation for chaos and disorganization, the Italian government—shortly after the unification of the country in 1861—ordered local officials to record the birth, marriage, and death of every citizen in each commune, or township.

J Since 1994, scientists at the University of Calabria have combed through these records in every one of Calabria's 409 *comuni* to compile an extraordinary survey. Coupling family histories with simple physiological[8] measurements of frailty and the latest genomic technologies, they set out to address fundamental questions about longevity. How much of it is determined by genetics? How much by the environment? And how do these factors interact to promote longevity—or, conversely, to hasten the aging process? To answer all those questions, scientists must start with rock-solid demographic[9] data.

K "Here is the book from 1905," explained Marco Giordano, one of Giuseppe Passarino's young colleagues, opening a tall green ledger. He pointed to a record, in careful cursive, of the birth of Francesco D'Amato on March 3, 1905. "He died in 2007," Giordano noted, describing D'Amato as the central figure of an extensive genealogical tree. "We can **reconstruct** the pedigrees[10] of families from these records."

L Cross-checking the ledger entries against meticulously detailed registry cards (pink for women, white for men)

[7] A **malady** is a disease or health problem.
[8] If something is **physiological**, it has to do with how the body functions.
[9] **Demographic** information is information about different groups of people in a particular society.
[10] Someone's **pedigree** is his or her background or ancestry.

going back to the 19th century, Giordano—along with researchers Alberto Montesanto and Cinzia Martino—has reconstructed extensive family trees of 202 nonagenarians[11] and centenarians in Calabria. The records document not only siblings of people who lived to 100, but also the spouses of siblings, which has allowed Passarino's group to do a kind of historical experiment on longevity. "We compared the ages of D'Amato's brothers and sisters to the ages of their spouses," Giordano explained. "So they had the same environment. They ate the same food. They used the same medicines. They came from the same culture. But they did not have the same genes." In a 2011 paper, the Calabrian researchers reported a surprising conclusion: Although the parents and siblings of people who lived to at least 90 also lived longer than the general population—a finding in line with earlier research—the **genetic** factors involved seemed to benefit males more than females.

M The Calabrian results on gender offer yet another hint that the genetic twists and turns that confer longevity may be unusually complex. Major European studies had previously reported that women are much likelier to live to 100, **outnumbering** male centenarians by a **ratio** of four or five to one, with the **implication** that some of the reasons are genetic. But by teasing out details from family trees, the Calabrian researchers discovered an intriguing paradox: The genetic **component** of longevity appears to be stronger in males—but women may take better advantage of external factors such as diet and medical care than men do.

N In the dimly lit, chilly hallway outside Passarino's university office stand several freezers full of tubes containing centenarian blood. The DNA from this blood and other tissue samples has revealed additional information about the Calabrian group. For example, people who live into their 90s and beyond tend to possess a particular version, or allele, of a gene important to taste and digestion. This allele not only gives people a taste for bitter foods like broccoli and field greens, which are typically rich in compounds that promote cellular health, but also allows cells in the intestine to extract nutrients more efficiently from food as it's being digested.

O Passarino has also found in his centenarians a revved-up[12] version of a gene for what is called an uncoupling protein. The protein plays a central role in metabolism—the way a person consumes energy and regulates body heat—which in turn affects the rate of aging.

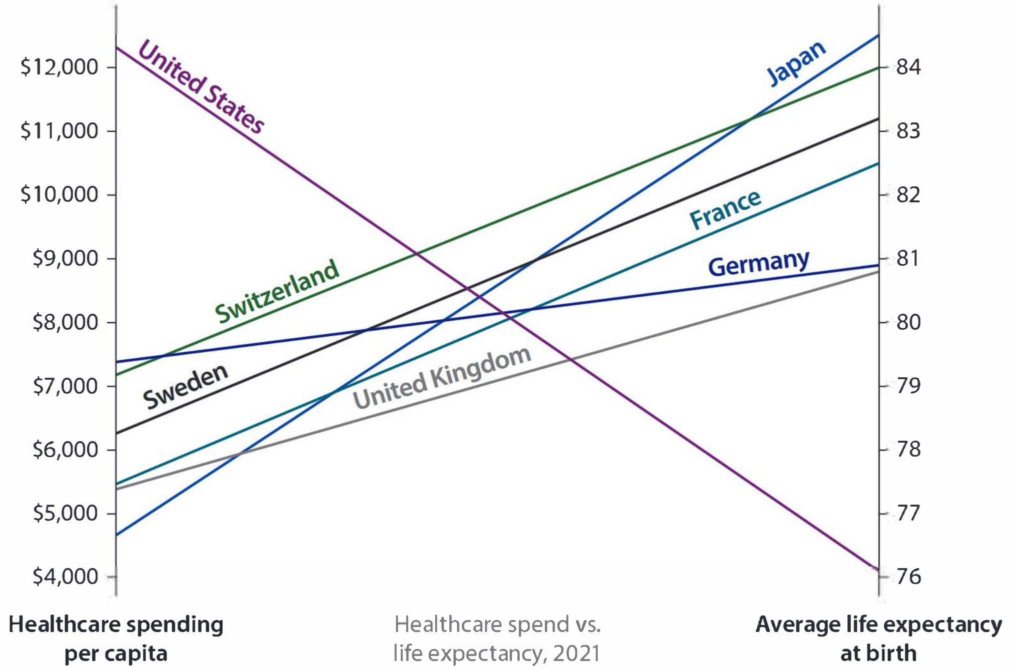
P "We have dissected five or six pathways that most influence longevity," says Passarino. "Most of them involve the response to stress, the metabolism of nutrients, or metabolism in general—the storage and use of energy." His group is currently examining how environmental influences—everything from childhood diet to how long a person attends school—might modify the activity of genes in a way that either promotes or curtails longevity.

Q Around the world, studies are being done to determine the causes of longevity and health in old age. If nothing else, the plethora of new studies indicates that longevity researchers are pushing the scientific conversation to a new level. In October 2011, the Archon Genomics X Prize launched a race among research teams to sequence the DNA of a hundred centenarians (dubbing the contest "100 over 100").

[11] A **nonagenarian** is someone who is 90 to 99 years old.
[12] If something is **revved up**, it is more active or intense than usual.

THE COST OF CARE

Are the funds used by countries on healthcare well spent? One way to ÿnd out is to compare per-person healthcare costs against longevity.



R But genes alone are unlikely to explain all the secrets of longevity, and experts see a cautionary tale[13] in recent results concerning caloric restriction. Experiments on 41 different genetic models of mice, for example, have shown that restricting food intake produces **outcomes** that are wildly **contradictory**. About half the mouse species lived longer, but just as many lived less time on a restricted diet than they would have on a normal diet. And last August, a long-running National Institute on Aging experiment on primates concluded that monkeys kept on a restricted-calorie diet for 25 years showed no longevity advantage. Passarino made the point while driving back to his laboratory after visiting the centenarians in Molochio. "It's not that there are good genes and bad genes," he said. "It's certain genes at certain times. And in the end, genes probably account for only 25 percent of longevity. It's the environment, too, but that doesn't explain all of it either. And don't forget chance."

S Which brought to mind Salvatore Caruso of Molochio, 107 years old and still going strong. Because he broke his leg 88 years ago, he was unfit to serve in the Italian Army when his entire unit was recalled during World War II. "They were all sent to the Russian front," he said, "and not a single one of them came back." It's another reminder that although molecules and mechanisms yet unfathomed[14] may someday lead to drugs that help us reach a ripe and healthy old age, a little luck doesn't hurt either.

Adapted from "On Beyond 100," by Stephen S. Hall: National Geographic Magazine, May 2013

Award-winning writer Stephen S. Hall has written extensively about science and society for nearly 30 years. His work has appeared in numerous publications including *The New York Times Magazine*.

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[13] A **cautionary tale** is a story or situation that can serve as a warning to people.
[14] If something is **unfathomed**, it has not been understood or explained, usually because it is strange or complicated.

UNDERSTANDING THE READING

A **UNDERSTAND MAIN IDEAS** Check (✓) the answers to the questions below. More than one answer may be possible.

1. Why is Calabria a good place to study longevity?
   - ☐ a. It has the best genetic research labs in Europe.
   - ☐ b. Many universities with good science departments are located there.
   - ☐ c. It has an unusual number of people over 90 and even over 100.
   - ☐ d. Many of the communities keep excellent family records.

2. What are some of the main points scientists have learned about longevity?

   In general, . . .
   - ☐ a. eating less increases longevity.
   - ☐ b. genetics contribute more to longevity in men than women.
   - ☐ c. women live longer because they look after themselves better.
   - ☐ d. how our bodies respond to stress and different foods affects our longevity.
   - ☐ e. our genes are the most significant factor affecting our longevity.

B **UNDERSTAND DETAILS** Work with a partner. Find information in the reading passage to answer these questions.

1. Why did the researchers compare centenarians with their spouses and siblings?

   ____

   ____

2. What resources did the researchers use to make family trees of people in Calabria?

   ____

   ____

An 89-year-old fisherman from Okinawa Island, Japan, shows off his muscles.